

VARUN VADDI

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SUMMARY

Data Science graduate with **3+ years of experience** building **end-to-end machine learning solutions** from data pipelines to production. Proven ability to **translate business requirements into predictive models**, analyze large structured & unstructured datasets, and communicate insights to technical and non-technical stakeholders. Experience delivering **measurable business impact** use-cases.

WORK EXPERIENCE

Data Science Intern, Wild Genomics, CA

May 2025 - Aug 2025

- Partnered with research scientists to address low-quality sequencing data blocking critical insights, designed automated **Python** pipeline with filtering & QC that increased **analysis-ready data from 30% to 80%**, enabling accurate predictive modeling for conservation decisions.
- Reduced **data redundancy by ~90%** through similarity-based clustering (97-99% identity), improving feature reliability for downstream ML models and accelerating time-to-insight for cross-sample biodiversity analysis.
- Enhanced species **classification accuracy from 85% to 95%** using optimized BLAST similarity thresholds, enabling genus-level precision that supported grant funding decisions and improved research outcome predictions.

Data Engineer, University of Houston, TX

Jul 2024 - Dec 2025

- Partnered with admissions leadership to support in developing of a predictive model **analyzing 40K+ applications per cycle**, identifying key graduation predictors, translated findings into automated admission thresholds that reduced **manual processing by 70%**, accelerated **decisions by 5 days**, and improved predicted **graduate success rates by 12%**.
- Built automated data pipelines integrating multiple admissions systems and designed Tableau dashboards enabling officers to **prioritize high-risk applications**, improving decision consistency & supporting evidence-based policy changes.

Software Engineer, Accenture, Hyderabad, India

Sep 2021 - Dec 2023

- Partnered with sales leadership to optimize SaaS data systems **managing 80M+ records**, streamlined analytics workflows & improved **deployment speed by 30%**, enabling faster identification of high-value leads and sales strategies.
- Led CRM data migration projects and developed SQL-based analytics pipelines that identified **conversion bottlenecks and optimized sales funnels**, directly **contributing to \$25M in revenue growth** through improved lead prioritization.
- Defined KPIs with business stakeholders and automated Tableau dashboards monitoring feature adoption and deal conversions, delivering actionable insights that **increased conversion rates by 10%** and informed product roadmap decisions.

PROJECTS

Urban Mobility Analytics Pipeline

- Built end-to-end analytics pipeline (Snowflake, dbt, Airflow, Kafka) addressing slow feature preparation for **8.6M+ taxi trips**, reduced **prep time by 95%** with 37 validations, enabling real-time demand forecasting to **optimize driver allocation** & reduce wait times.
- Analyzed **\$180M+ quarterly revenue** patterns using SQL and Tableau, uncovering **30% peak-hour demand lift** and **18% credit-card revenue premium** insights assisted in informed business strategies for **dynamic pricing optimization**.

Ads Policy Compliance Assistant

- Developed hybrid RAG system (BM25, embeddings, FAISS) to address 5-10 min manual ad review delays, **reduced review latency to <1 sec** while achieving **80% accuracy and 78% Recall@5**, enabling scalable **compliance for 1,000+ ads/sec** and reducing costs.
- Implemented structured LLM outputs with confidence scoring to eliminate inconsistent policy decisions, achieving **95% policy coverage** and enabling real-time risk detection that **reduced compliance violations by 40%**.

Named Entity Recognition on X posts

- Fine-tuned a **BERT-based NER model** on **11K social media posts**, improving macro-F1 to **89% across 7 entity types** and enabling faster detection of emerging trends for marketing analytics.
- Optimized **sub-word token alignment and BIO tag reconciliation**, reducing entity extraction errors by **12%** and supporting scalable social listening pipelines processing **100K+ daily posts**.

EDUCATION

University of Houston

Master of Science (MS), Engineering Data Science | GPA: 3.7/4

TX, USA

Dec 2025

SKILLS

Programming Languages: Python, SQL, PySpark, R, RBMS.

Tools/Frameworks: NumPy, SciPy, Pandas, Seaborn, Matplotlib, Plotly, PyTorch, TensorFlow, Scikit-learn, Git, CI/CD, Jira.

Data Eng: Snowflake, Databricks, Apache Spark, DBT, Airflow, Docker, Kafka, Kubernetes, Oracle, Tableau, Power BI, Excel.

Competencies: AWS, Azure, Machine Learning, Neural Networks, Deep Learning, NLP (Gen AI, LLM, RAG, Transformers), Predictive Modeling, Statistical Analysis, Feature Engineering, Hyperparameter tuning, A/B Testing, Inferential Methods, Causal Inference.